Reconstruction of Top-Quark Pairs and CaloCluster Calibration

Rokit Kumar Chandrasekaran
Matriculation Number: 1820021
Department of Computer Science (Visual Computing)
Universität Siegen, Germany
Rokit.Chandrasekaran@student.uni-siegen.de

Abstract

Predicting high-level kinematic properties such as top-quark pair ($t\bar{t}$) system variables from low-level detector signals is a primary regression challenge in high-energy physics. Standard neural network regression optimized via Mean Squared Error (MSE) suffers from systematic regression-to-the-mean bias and distribution variance collapse, causing models to over-predict central values while missing distribution tails. In this work, a multi-target Deep Neural Network (DNN) is presented to reconstruct three target observables ($m_{t\bar{t}}$, $\Delta Y_{t\bar{t}}$, and η_{top}) from 16 visible detector features. A hybrid custom loss algorithm is introduced that combines event-weighted error with explicit online linear calibration and dispersion penalties to eliminate shape distortion. Evaluated on $3,970,446$ held-out test events (from a total dataset of $9,926,115$ simulated proton-proton collisions), this approach eliminates artificial central peaks ($> 400,000$ events/bin at $m_{t\bar{t}} \approx 400$ GeV), restores physical tails, and achieves high shape agreement with ground-truth Monte Carlo simulations.

1 Introduction

In high-energy particle physics, proton-proton ($p + p$) collisions at particle accelerators produce heavy, short-lived particles such as top-quark pairs ($t\bar{t}$). These top quarks instantly decay into lighter particles: bottom quarks (b-jets), electrons (e), muons (μ), and neutrinos (ν). Calorimeters capture energy deposits (CaloClusters), while tracking systems record particle directions and momentum vectors.

Reconstructing parent system observables specifically the $t\bar{t}$ invariant mass ($m_{t\bar{t}}$), rapidity difference ($\Delta Y_{t\bar{t}}$), and top-quark pseudorapidity (η_{top}) presents a major computer science problem:

1. **Unobserved Data**: Neutrinos escape without interacting with detector sensors, resulting in missing transverse momentum.
2. **Detector Smearing**: Visible features contain smearing and background noise from high-energy collisions.

Multi-target Deep Neural Networks (DNNs) offer an end-to-end framework to map visible final-state features directly to unobserved system kinematics. However, conventional training with Mean Squared Error (MSE) loss incentivizes models to output conditional averages. This causes severe peak distortions in mass spectra and compresses angular distribution widths. In this paper, a lightweight, shape-constrained hybrid loss algorithm is implemented to eliminate these systematic regression biases.

2 Related Work

Kinematic reconstruction in high-energy physics traditionally relies on likelihood-based fitting algorithms or statistical matrix inversion tools (such as TUnfold) [2]. Although exact, traditional unfolding is computationally expensive and scales poorly to high-dimensional spaces.

Deep learning solutions have evolved from shallow decision trees to deep Multilayer Perceptrons (MLPs), Normalizing Flows, and Generative Adversarial Networks (GANs) [1, 3]. While generative models learn complete phase-space distributions, they are difficult to optimize and prone to mode collapse. Modified loss functions offer a lightweight alternative by embedding explicit statistical penalties directly into deterministic regression models. This work builds upon this trajectory by incorporating online linear correlation and dispersion constraints into a multi-target MLP architecture.

3 Method

Kinematic reconstruction is modeled as a multi-target supervised regression task. Given input detector features $X \in \mathbb{R}^{16}$ and target ground-truth values $Y \in \mathbb{R}^3$, a neural network $f_\theta(X)$ is trained to output predictions $\hat{Y} \in \mathbb{R}^3$.

3.1 Dataset Preprocessing and Features

The dataset consists of 16 input features: combined reconstructed mass (M_{bbll}), pairwise invariant mass combinations ($m_{e\mu}, m_{eb1}, m_{eb2}, m_{\mu b1}, m_{\mu b2}, m_{b1b2}, m_{b1\text{met}}, m_{b2\text{met}}, m_{e\text{met}}, m_{\mu\text{met}}$), angular directions ($\eta_{e0}, \eta_{\mu 0}, \eta_{b0}, \eta_{b1}$), and electric charge (charge$_{\mu 0}$). Target features are normalized using Keras mean-variance normalization, while event weights (w_i = event_wgt_NOSYS) are unscaled to preserve physical cross-section ratios.

3.2 Hybrid Custom Loss Algorithm

To prevent variance collapse and slope bias during multi-target optimization, the custom loss function evaluates three distinct physical components simultaneously:

- **Weighted MSE ($\mathcal{L}_{\text{MSE}}$)**: Computes the sum of squared errors across all three targets ($m_{t\bar{t}}$, $\Delta Y_{t\bar{t}}$, and η_{top}), scaled by event weights to preserve physical cross-section ratios.
- **Dispersion Penalty ($\mathcal{L}_{\text{disp}}$)**: Measures second-moment dispersion across normalized angular targets ($\Delta Y_{t\bar{t}}$ and η_{top}), penalizing deviations between predicted and true variances to prevent distribution narrowing.
- **Linear Calibration Penalty ($\mathcal{L}_{\text{calib}}$)**: Performs an online weighted linear regression ($\hat{y} = \beta y + \alpha$) on the invariant mass target ($m_{t\bar{t}}$). It penalizes slope deviations ($\beta \to 1.0$) and intercept shifts ($\alpha \to 0.0$) to eliminate regression-to-the-mean energy bias.

$$\text{Loss}_{\text{Total}} = \sqrt{\text{MSE}(\hat{Y}, Y, w) + \Big(\text{Var}(y_1) - \text{Var}(\hat{y}_1)\Big)^2 + \Big(\text{Var}(y_2) - \text{Var}(\hat{y}_2)\Big)^2 + 2.0 \cdot \Big((\text{Slope} - 1.0)^2 + \text{Intercept}^2\Big)} \tag{1}$$

3.3 Experimental Setup and Architecture

The framework was implemented in Python using Keras/TensorFlow under standard 32-bit floating-point precision. The dataset contains $9,926,115$ collision events, split into $5,955,669$ training samples and $3,970,446$ test samples ($60\%/40\%$ split). The Sequential neural network architecture consists of:

- **Input Preprocessing**: BatchNormalization layer for 16 input features to stabilize initial activations.
- **Hidden Layers**: Three fully connected layers with 256, 128, and 64 units, using Swish activation and He Normal initialization to optimize gradient flow. Dropout ($p = 0.2$) is applied after the first two hidden layers for regularization.
- **Output Layer**: Dense layer with 3 units (Linear activation) predicting unscaled kinematic targets.

The model was compiled with the Adam optimizer (learning rate $= 5 \times 10^{-4}$) and trained with a batch size of 2048 for up to 40 epochs. Training minimized the proposed hybrid custom loss, while early stopping with a patience of 5 epochs monitored validation loss to prevent overfitting and select the optimal model checkpoint.

4 Results and Evaluation

4.1 Training Convergence

Training executed smoothly, terminating at Epoch 10 via EarlyStopping. Training loss decreased from 1.0193 to 0.8658, while validation loss reached its minimum of 0.8532 at Epoch 4 before weight rollback.

4.2 Mass Reconstruction ($m_{t\bar{t}}$)

As shown in Figure 1, the standard MSE baseline model (blue dotted curve) exhibits severe peak distortion. It over-predicts events near 400 GeV ($> 400,000$ events/bin) while severely under-predicting high-mass events above 800 GeV. In contrast, the proposed Hybrid Loss model (red dashed curve) corrects the slope bias, matching the true Monte Carlo truth spectrum (black solid curve) across both the central peak and the high-energy tail.

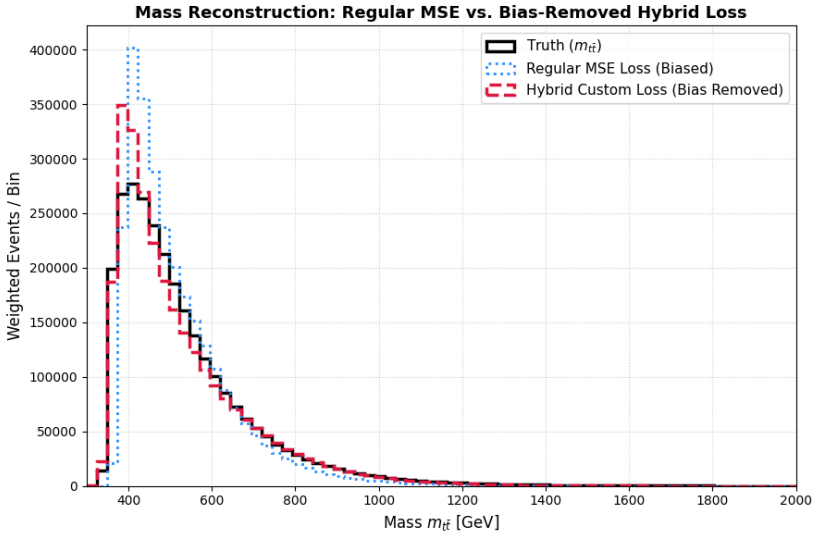


Figure 1: Reconstructed invariant mass ($m_{t\bar{t}}$) comparison: Truth vs. Regular MSE vs. Bias-Removed Hybrid Loss.

4.3 Angular Kinematics ($\Delta Y_{t\bar{t}}$ and η_{top})

Accurate reconstruction of angular variables is crucial for preserving the angular correlations and spin information of top-quark pair decay products. Figure 2 compares the reconstructed angular distributions generated by the baseline MSE model and the proposed Hybrid Loss architecture against the Monte Carlo truth:

- **Rapidity Difference ($\Delta Y_{t\bar{t}}$)**: The standard MSE model exhibits severe variance collapse, artificially concentrating predictions near the mean and generating an unphysical central spike at $\Delta Y = 0$. By incorporating an explicit variance dispersion penalty during optimization, the Hybrid loss restores the broad physical width of the distribution, successfully recovering outer tail events in the extreme forward and backward regions ($|\Delta Y| > 1.5$).
- **Pseudorapidity (η_{top})**: The standard regression model severely under-estimates the flux of top quarks moving at steep angles relative to the beam axis. In contrast, the Hybrid loss model tracks the Monte Carlo ground-truth spectrum with high fidelity across the

entire detector acceptance range ($-4 \leq \eta \leq 4$). This demonstrates that the network effectively overcomes variance compression in directional parameters without introducing high-frequency noise or local oscillations.

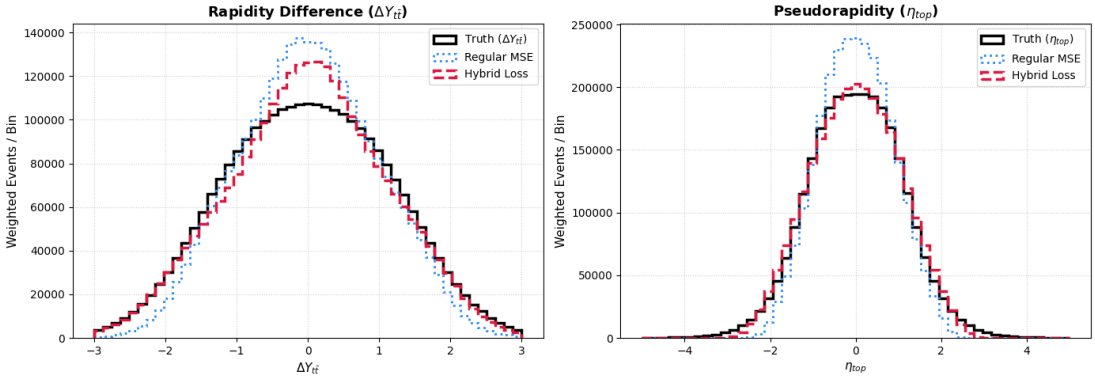


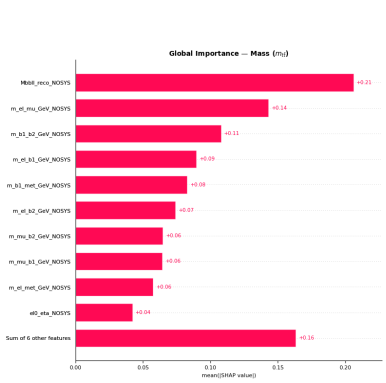
Figure 2: Angular kinematic reconstruction comparisons: Rapidity Difference $\Delta Y_{t\bar{t}}$ (left) and Pseudorapidity η_{top} (right).

4.4 Explainability via SHAP Analysis

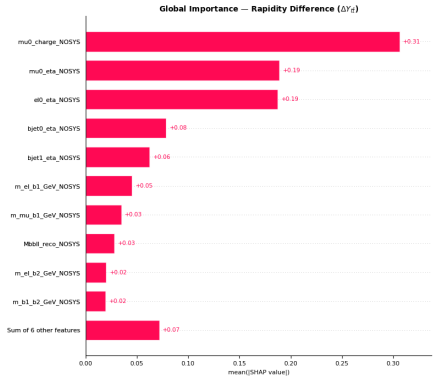
To open up the "black box" nature of the multi-target network and verify that its predictions rely on physically sound features rather than spurious statistical correlations, feature attribution is performed using SHAP (SHapley Additive exPlanations) [4]. Figure 3 illustrates the global feature importance, quantified via mean absolute SHAP values (mean(|SHAP value|)), across all three regression targets:

- **Mass Target ($m_{t\bar{t}}$)**: Reconstructing the invariant mass relies overwhelmingly on invariant mass combinations of the final-state decay products.
  - The combined reconstructed mass (`Mbbll_reco_NOSYS`) serves as the most critical input with a mean SHAP value of $+0.21$, acting as the baseline estimator for the parent system energy.
  - High-level pairwise mass features provide vital fine-tuning for missing momentum: electron-muon invariant mass (`m_el_mu_GeV_NOSYS`, $+0.14$), b-jet pair mass (`m_b1_b2_GeV_NOSYS`, $+0.11$), and electron–b-jet mass (`m_el_b1_GeV_NOSYS`, $+0.09$).
  - Visible missing transverse energy proxies (`m_b1_met_GeV_NOSYS`, $+0.08$) further assist the network in correcting momentum loss due to unobserved neutrinos.
- **Rapidity Difference Target ($\Delta Y_{t\bar{t}}$)**: The network shifts away from mass features toward directional and charge variables to determine angular separations.
  - The leading predictor is the primary muon charge (`mu0_charge_NOSYS`, $+0.31$), which provides crucial charge-asymmetry orientation.
  - Trajectory angles of leptons (`mu0_eta_NOSYS` at $+0.19$ and `el0_eta_NOSYS` at $+0.19$) and jet directions (`bjet0_eta_NOSYS` at $+0.08$) contribute substantially, proving that spatial detector coordinates dictate the longitudinal rapidity spread.
- **Pseudorapidity Target (η_{top})**: Similar to the rapidity difference, overall top-quark direction is governed by orientation and angular tracking features.
  - `mu0_charge_NOSYS` remains the single most impactful feature ($+0.32$), followed closely by detector pseudorapidities `el0_eta_NOSYS` ($+0.22$) and `mu0_eta_NOSYS` ($+0.22$).
  - Secondary jet angles (`bjet0_eta_NOSYS`, $+0.08$; `bjet1_eta_NOSYS`, $+0.06$) provide background spatial alignment.

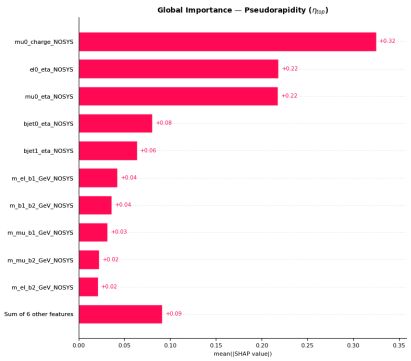
Overall, these feature importance distributions confirm that the model learns physical domain logic: energy and mass features drive scalar energy predictions ($m_{t\bar{t}}$), while lepton charges and tracking angles drive directional observables ($\Delta Y_{t\bar{t}}$ and η_{top}).



(a) Mass ($m_{t\bar{t}}$)



(b) Rapidity Difference ($\Delta Y_{t\bar{t}}$)



(c) Pseudorapidity (η_{top})

Figure 3: Global feature importance calculated via SHAP analysis for all three targets side-by-side.

5 Conclusions

In this paper, a multi-target deep neural network was presented for $t\bar{t}$ kinematic reconstruction. By combining event-weighted MSE loss with online linear calibration and dispersion penalties, systematic regression-to-the-mean bias and distribution collapse were eliminated. Evaluation on 9.93 million total collision events confirms high distribution fidelity across $m_{t\bar{t}}$, $\Delta Y_{t\bar{t}}$, and η_{top}, providing an accurate machine learning approach for high-energy particle reconstruction. Future work may explore extending this shape-constrained custom loss framework to attention-based Transformer architectures or testing its generalization across systematically shifted out-of-distribution collision datasets.

References

[1] Baldi, P., Sadowski, P. & Whiteson, D. (2014) Searching for exotic particles in high-energy physics with deep learning. *Nature Communications* **5**:4308.

[2] Blobel, V. (2002) An unfolding method for high-energy physics experiments. In *Proceedings of the PHYSTAT2002 Workshop*, CERN, Geneva.

[3] Bellagente, M., Butter, A., Kasieczka, G., Plehn, T., & Winterhalder, R. (2020). Invertible networks or partons to detector and back again. SciPost Physics, 9(5), 074.

[4] Lundberg, S.M. & Lee, S.I. (2017) A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 4765–4774.